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Tectonic rotations in the Cascade Range of southern Washington

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ABSTRACT

Paleomagnetic directions from 34 widely distributed sampling sites in presumed Oligocene volcanic rocks from the Cascade Mountains of southern Washington are well grouped with a mean and confidence interval of declination = 27°, inclination = 64°, and α_{95} = 4.5°. When compared to an expected direction, computed from the Oligocene reference pole for stable North America, a clockwise rotation of 33.5° ± 14° has apparently occurred. This result is not significantly different from several other studies from neighboring parts of the Cascades and Coast Range of southern Washington and suggests that the two ranges have rotated as a single unit during post-Eocene time. Comparison with the results from rocks of similar age in Oregon suggests that at least two crustal blocks are involved in Pacific Northwest tectonics. Inclinations in most Coast Range and Cascade Paleogene rocks indicate that these provinces have been translated slightly northward, as well as rotated.

INTRODUCTION

In recent years the Pacific Northwest Coast Ranges have been the focus of several paleomagnetic investigations (Fig. 1; Table 1). Studies by Simpson and Cox (1977) and Magill and others (1981) on rocks of Eocene age and by Beck and Plumley (1980) on Eocene, Oligocene, and Miocene intrusive rocks, all within the Oregon Coast Range, show that the 350-km-long range is a single allochthonous block that has rotated, relative to the North American interior, by as much as 75° clockwise since Eocene time.

Work on Eocene rocks within the Washington Coast Range (Globerman and Beck, 1979; Wells and Coe, 1979) also indicates a clockwise rotation, but probably less than in Oregon (Fig. 1). Globerman and Beck (1979) reported about 36° of rotation, whereas the Wells and Coe (1979) study shows at most 10°. On the basis of these and other data, Beck and Plumley (1980) suggested that there are at least two crustal blocks involved in the rotation. They also suggested that part of the boundary between the blocks may coincide with the present mouth of the Columbia River.

Considering the apparently allochthonous nature of the Coast Ranges, it seems appropriate to ask if the volcanic Cascade Range to the east also was involved in the rotation. Paleomagnetic data reported by Beck (1962) suggest that the southern tip of the Cascade Range in northern California probably has rotated clockwise by about 13°. Beck and Burr (1979) reported about 25° of clockwise rotation from the Goble Volcanic Series of mid-Eocene to mid-Oligocene age in southern Washington. The Goble Volcanics occur at the base of the Lower Western Cascade Group of southern Washington (Hammond, 1979, Fig. 4). Work by Magill and Cox (1981) in the Oregon Cascades also indicates a significant amount of clockwise rotation (Fig. 1,

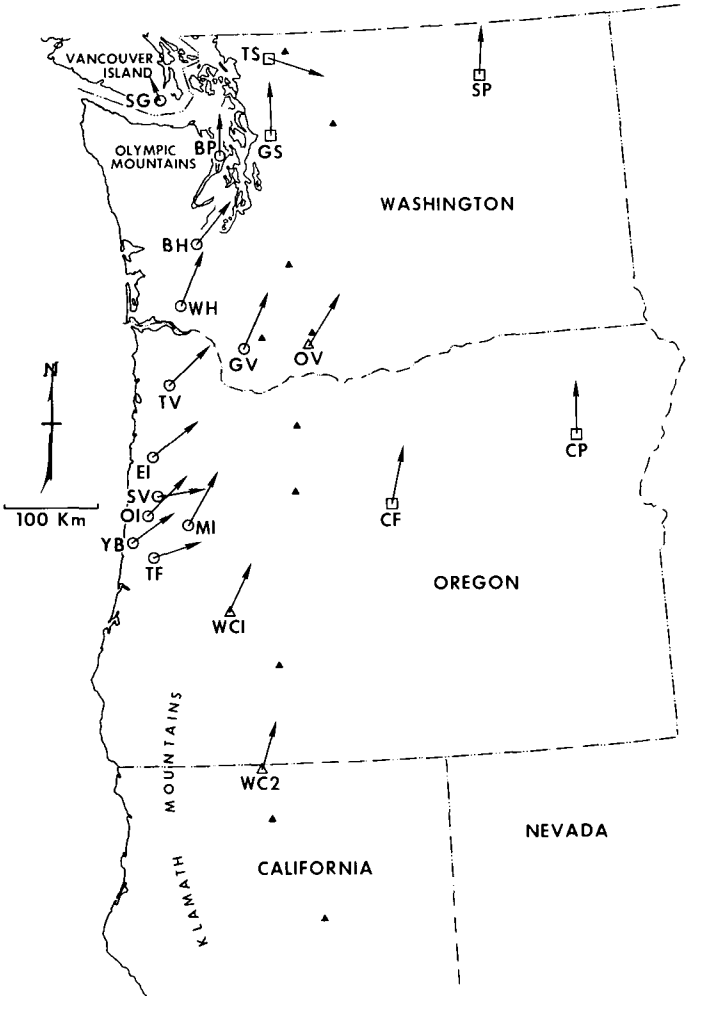


Figure 1. Locations of paleomagnetic studies of Tertiary rocks in Pacific Northwest. Arrows indicate amount of rotation, defined as difference between expected and observed magnetic declination (Beck, 1980). Arrow pointing due north indicates no rotation; thus, symbol for SG (southern Vancouver Island) shows that magnetic vector for East Sooke Gabbro is deflected 22.5° westward of its reference direction. Note preponderance of eastward deflections. Open circles indicate studies on Tertiary rocks of Oregon and Washington Coast Ranges. Open triangles denote Cascade arc rocks; squares indicate other studies. Solid triangles are major Cascade volcanoes. Abbreviations are explained in Table 1.

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WC1). Hammond (1979) suggested that the Cascades and the Coast Ranges lay adjacent to one another during at least part of the time that the Coast Ranges were rotating; Hammond's conclusions are based on stratigraphic relationships reported by Hoover (1963) and Wells and Peck (1961) in Oregon and by Snavely and others (1958) and Roberts (1958) in southwestern Washington.

Although evidence for Cascade rotation is mounting, there is a significant gap in both space and time between the Goble Volcanics (Beck and Burr, 1979) and the mid-Tertiary Oregon Cascade study of Magill and Cox (1981). In an attempt to fill a part of this gap, we instituted a paleomagnetic investigation of the Ohanapecosh Formation of southern Washington. We present here the results of that study.

TABLE 1. PALEOMAGNETIC STUDIES OF TERTIARY ROCKS FROM THE PACIFIC NORTHWEST

Study	Unit/Rock Type	Age	Reference
TS	Twin Sisters Dunite	Paleocene?	Beck (1975)
SP	Sanpoil Volcanics	Eocene	Schwimmer & others (1979)
SG	East Sooke Gabbro	Oligocene?	Symons (1973)
GS	Grotto/Snoqualmie Batholiths	Miocene	Beske & others (1973)
BP	Bremerton/Pt. Townsend Volcanics	Eocene?	Beck & Engebretson (1980)
BH	Black Hills Volcanics	Eocene	Globerman & Beck (1979)
WH	Willapa Hills Volcanics	Eocene	Wells & Coe (1979)
GV	Goble Volcanics	Oligocene	Beck & Burr (1979)
OV	Ohanapecosh Volcanics	Oligocene	This paper
TV	Tillamook Volcanics	Eocene?	Magill & others (1981)
CP	Columbia Plateau Basalts	Miocene	Watkins & Baski (1974)
EI	Eocene Basaltic Sills	Eocene	Beck & Plumley (1980)
SV	Siletz Volcanics	Eocene	Simpson & Cox (1977)
CF	Clarno Volcanics	Eocene	Beck & others (1978)
OI	Oligocene Basaltic Sills	Oligocene	Beck & Plumley (1980)
MI	Miocene Basaltic Sills	Miocene	Beck & Plumley (1980)
YB	Yachats Basalts	Eocene	Simpson & Cox (1977)
TF	Tyee-Flournoy sedimentary rocks	Eocene	Simpson & Cox (1977)
WC1	Western Cascade Series Volcanics	Oligo-Miocene	Magill & Cox (1981)
WC2	Western Cascade Series Volcanics	Oligo-Miocene	Beck (1962)

Note: Locations and rotations shown in Figure 1. Listed from north to south.

GEOLOGIC SETTING

The Ohanapecosh Formation can be traced from its type locality near Mt. Rainier National Park, Washington, southward about 110 km to the Wind River area of Skamania County, Washington, where most of this study was carried out (Fig. 2). In the area between Mt. Adams and Mt. St. Helens, Ohanapecosh rocks crop out only in a few places; they are covered by younger rocks from the two volcanoes (Wise, 1970; Simon, 1972). The Ohanapecosh Formation, described in detail by Fiske and others (1963), is a thick sequence of subaerial and subaqueous volcaniclastic rocks, mudflows, and lava flows. It makes up part of the lower Western Cascade Group (Hammond, 1979), and as such is among the oldest Cascade arc rocks exposed in southern Washington. Most such rocks in Washington appear to be approximately Oligocene in age. Ohanapecosh rocks have been dated, on the basis of plant fossils, as late Eocene (Fiske and others, 1963) at the type locality and, more recently, as early to middle Oligocene (31 to 37 m.y.) by J. Vance of the University of Washington (1979, personal commun.), by fission-track dating. Those in the Wind River area were assigned an Oligocene age by Wise (1970), on the basis of floral remains.

About 25 km west of the Wind River area, Schriener (1978) has informally divided the Skamania Volcanic Series of Trimble (1963) into the East Fork formation and the Skamania formation. The East Fork formation, which crops out mainly along the East Fork of the Lewis River, is a sequence of volcaniclastic rocks and basalt flows. The overlying Skamania formation is a series of basalt flows with minor clastic units. The ages of the East Fork and Skamania formations have not been determined precisely; however, lithologic and stratigraphic similarities and relationships allow tentative correlation with the Ohanapecosh formation and the assignment of a probable Oligocene age (Schriener, 1978).

Two periods of folding were recognized by Hammond (1979) in the Lower Western Cascade Group. An older, northwest-trending system is approximately contemporaneous with Ohanapecosh deposition and is represented by broad, open folds whose limbs dip 10° to 45°. A younger pattern, which is the more prominent of the two in the Wind River area, is an extension of the Yakima fold system and trends east-west. Dips are generally less than 40°.

All Ohanapecosh flows seem to have undergone some alteration. In most places the alteration is slight, but in others, particularly where the deeper stratigraphic levels of the Ohana-

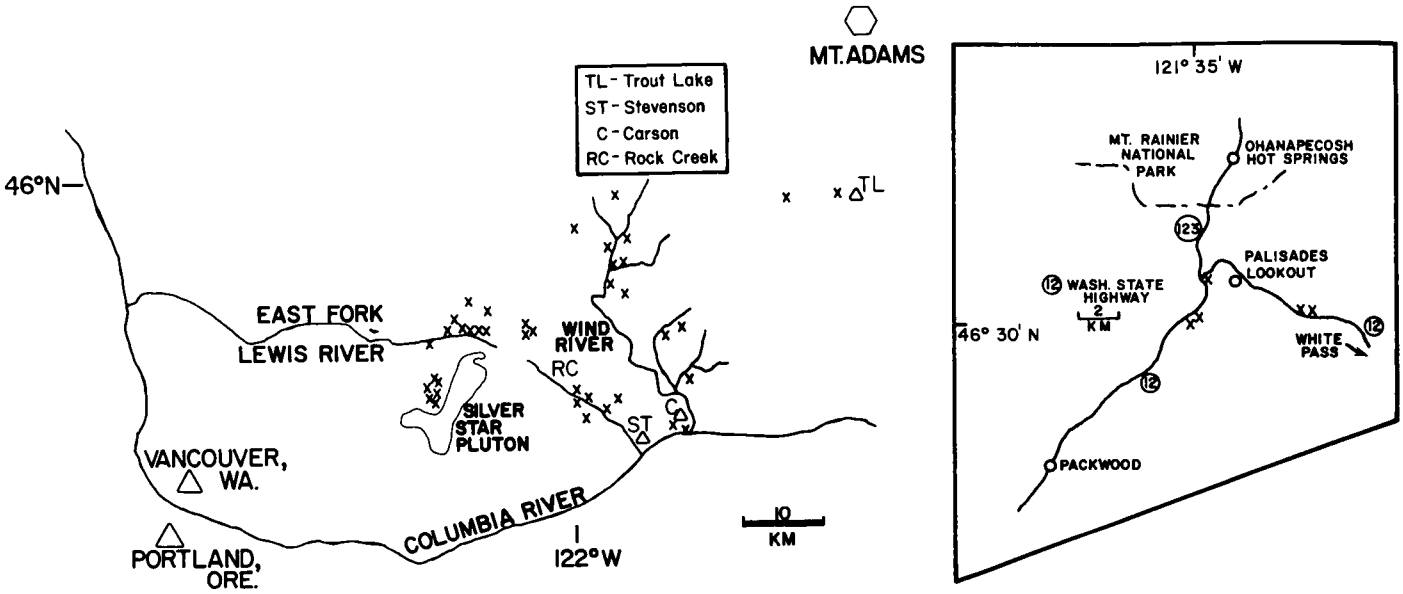


Figure 2. Map of part of southern Washington showing sampling locations in Wind River area, modified from Huntting and others (1961), and map of sampling locations near Mt. Rainier National Park.

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pecosh are exposed, alteration is extreme (Fiske and others, 1963). In the East Fork formation, alteration has in some places resulted in nearly total replacement of the mafic minerals by chlorite, clay, opaques, and epidote (Schriener, 1978). The alteration affects magnetic stability, as shown below.

PALEOMAGNETISM

Portable core-drilling equipment and in situ orienting devices, similar to those described by Doell and Cox (1965) (but including a sun compass) were used to collect a total of 336 samples from lava flows at 40 sites in southern Skamania County, Washington, and from five sites in the Ohanapecosh type section near Mt. Rainier, Washington (Fig. 2).

All samples were magnetically cleaned, by the alternating-field (A-F) method. In general, the magnetic stability of the rock was good; only one site was rejected because its circle of 95% confidence exceeded our arbitrary cutoff level of 15°. Five additional sites were rejected because they showed very erratic and unstable behavior during A-F demagnetization (Fig. 3). Rocks from the rejected sites tended to be the most severely altered. A more complete discussion of the behavior of the rejected sites during demagnetization may be found in Bates (1980). One site, whose mean direction gave a strong westerly declination and a shallow inclination, also was rejected, because it was clearly divergent from the tight grouping formed by the mean magnetic directions of the other sites (Fig. 4). This particular site was located in an isolated outcrop, which may have been slumped or otherwise displaced. Three sites, located in dikes intruding the Ohanapecosh, were rejected because their age, other than being post-Ohanapecosh, could not be determined independently and their directions of remanent magnetization were not statistically different from that of the present dipole field. Individual samples (32) that diverged from the site mean by more than twice the angular standard deviation also were rejected.

Both normal and reversed polarities of magnetization are present within the Ohanapecosh Formation and the East Fork and Skamania formations. The mean directions of magnetization of the 16 sites showing reversed polarity and the 18 sites of normal polarity are antiparallel at the 95% confidence level, indicating that any secondary components of magnetization have been substantially reduced.

The mean direction of remanent magnetization for the Ohanapecosh Formation has been determined from the remaining 228 samples representing 34 sites (Fig. 4). The 34 sites are distributed as follows: 4 at the Ohanapecosh type section, 19 in the Ohanapecosh Formation, Wind River area, and 11 in the East Fork and Skamania formations (Fig. 2). Results are summarized in Table 2.

All sites were corrected for tilt (dip observed in the field), to determine the original direction of magnetization. However, because of the tendency for lava flows to conform to the land surface, in principle a distinction should be made between original dip and postmagnetization tilting. Unfortunately, this distinction is in most places very difficult, if not impossible, to make. "Correcting" for original dip undoubtedly introduces scatter into the data set, but because this "correction" presumably is small and probably random, the mean direction is unlikely to be affected significantly. In fact, the scatter of mean directions was significantly reduced after application of tilt correction to all sites. The precision parameter of Fisher (1953) increased from 20.1 to 30.7, a statistically significant change (Cox, 1969), whereas the circle of 95% confidence decreased from 5.4° to 4.4°, verifying the presence of important postmagnetization tectonic tilting and constituting a positive fold test (McElhinny, 1973). The tilt correction made little difference in the overall mean. The mean directions of remanent magnetization, calculated with and without tilt correction, for the Ohanapecosh Formation are not statistically distinct at the 95% confidence level and differ by less than 7° of declination.

The mean direction of remanent magnetization for the Ohanapecosh (tilt corrected) is declination = 26.8° east, inclination = 63.6° down, $\alpha_{95} = 4.4°$, $\kappa = 30.6$ (Fig. 4). When compared to the expected direction of magnetization for the study area, calculated from the Oligocene pole of Irving (1979), a discordance in declination of 33.5° ± 14° clockwise is noted. Because inclination is not significantly different from that calculated

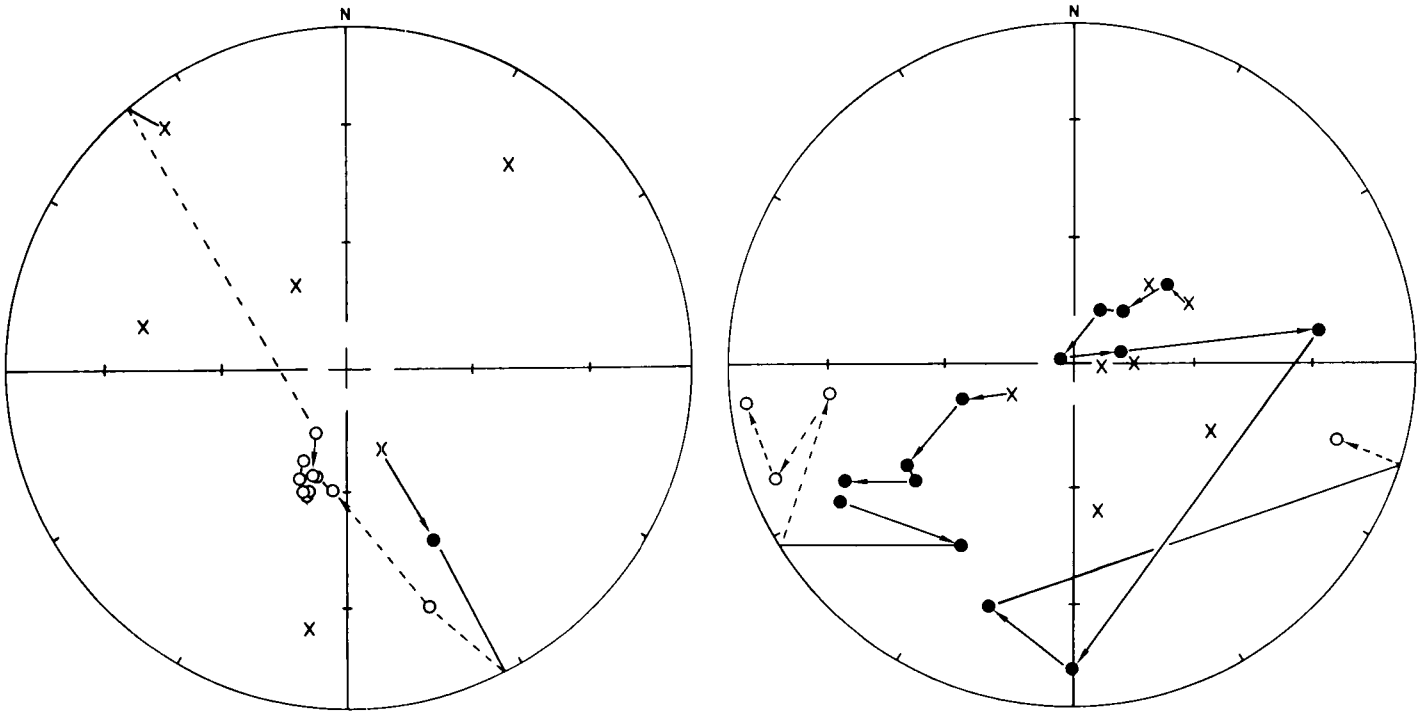


Figure 3. Lower-hemisphere, equal-area projections showing effects of A-F demagnetization on stable (a) and unstable (b) rocks. Solid circles indicate normal polarity; open circles indicate reversed polarity; x indicates natural remanent magnetization direction; arrows show path of stepwise demagnetization.

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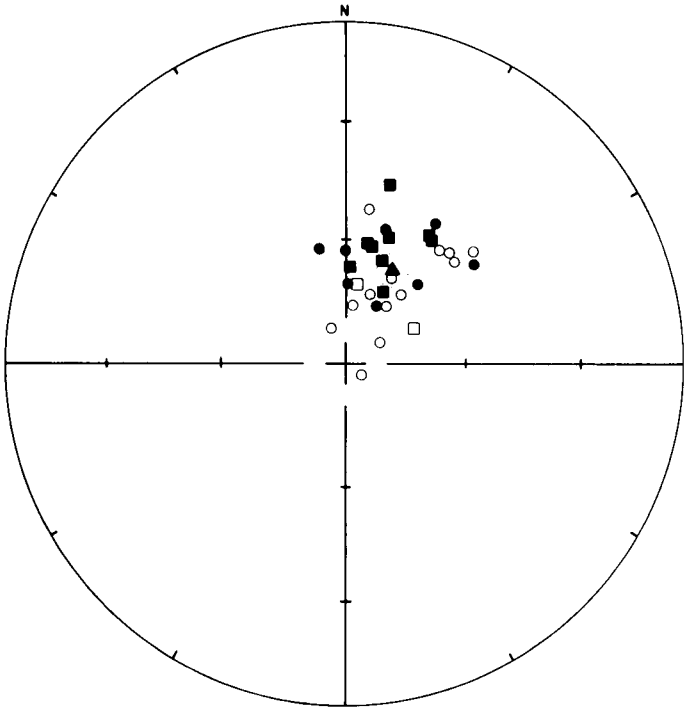


Figure 4. Site-mean directions of remanent magnetization for Ohanapecosh Formation and East Fork and Skamania formations. Lower-hemisphere equal-area plot. Solid symbols indicate normal polarity; open symbols indicate reversed polarity inverted through center of projection. Triangle indicates mean of site means (with circle of 95% confidence). Circles indicate Ohanapecosh Formation; squares indicate East Fork and Skamania formations.

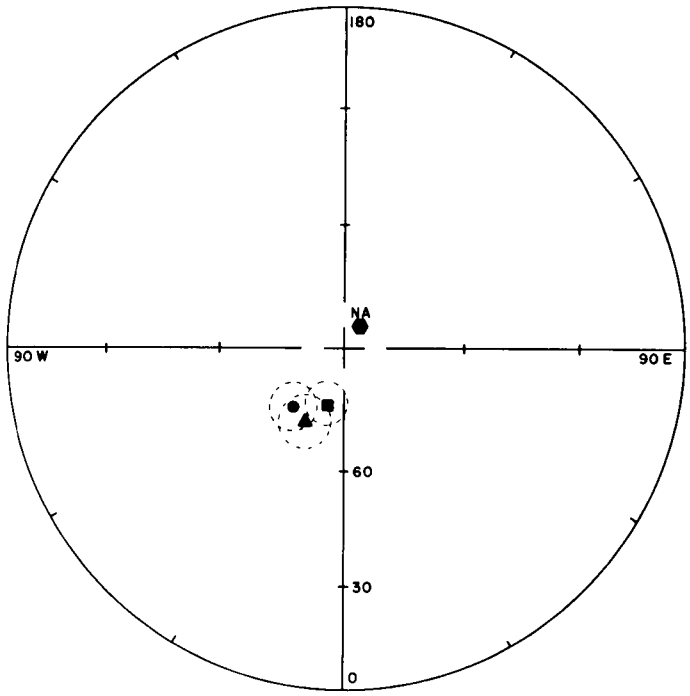


Figure 5. Paleomagnetic poles and 95% confidence intervals for Goble Volcanic Series (square); Ohanapecosh Formation, East Fork and Skamania formations (circle); and Black Hills (triangle). NA = early Oligocene reference pole for North America (Irving, 1979).

from the Oligocene pole, it appears that the Ohanapecosh rocks did not move significantly north or south, relative to stable North America, but rather rotated more or less in place about a vertical axis (however, see below). Translations along latitude lines (rotation about the geographic pole) also are possible, because these cannot be detected by paleomagnetic methods.

It has been suggested that some of our sites, particularly East Fork and Skamania rocks, may be as young as late Oligocene–early Miocene and, therefore, are significantly younger than the rest of the rocks sampled in this study (P. E. Hammond, 1980, personal commun.). To examine this possibility, we have separated the data set into two groups, a "young" group comprising the East Fork and Skamania rocks and a single site from the Wind River area, and an "old" group made up of all remaining sites. Table 3 shows average directions of magnetization for these subsets. The mean declination for the "young" groups is some 8° nearer the expected direction than is the mean for the "old" group; however, the two directions are not significantly different at the 95% confidence level. Either the two groups are not greatly different in age, or little rotation occurred during the Oligocene. Further details of the laboratory methods and within-site statistics are available in Bates (1980).

TABLE 2. PALEOMAGNETIC RESULTS FROM EARLIEST VOLCANIC ROCKS CASCADE RANGE, SOUTHERN WASHINGTON

Area	N	$\bar{D}$ (°)	$\bar{I}$ (°)	R	α_{95}	Polarity
Ohanapecosh Fm., type section	4	11.0	65.5	3.71	30.0	Mixed (3N, 1R)
Ohanapecosh Fm., Wind River area	19	33.5	63.0	17.91	8.5	Mixed (6N, 13R)
East Fork-Skamania fms.	11	22.0	61.5	10.80	6.5	Mixed (9N, 2R)
All flows	34	27.0	63.5	32.92	4.5	
Expected direction		353.5	65.0			

Note: N, number of flows; $\bar{D}$, $\bar{I}$, mean declination and inclination; R, length of vector sum (McElhinny, 1973); α_{95}, circle of 95% confidence. Expected direction calculation from the reference poles of Irving (1979).

INTERPRETATION

Our results suggest that at least the southernmost part of the Cascade Mountains in Washington, and possibly all of the Washington Cascades south of Mt. Rainier, are displaced relative to the stable North American interior. Additionally, the results reported by Beck and Burr (1979) from the Goble Volcanic Series, immediately west of the study area, and those of Globerman and Beck (1979) from the area to the northwest in the Black Hills of the Washington Coast Range show post-Eocene rotations that are statistically indistinguishable from our results. This suggests that the Washington Coast Range and the southern Cascades of Washington rotated together as a single unit (Fig. 5), as proposed previously by Hammond (1979). This large crustal

TABLE 3. SEPARATION OF SITES BY AGE

Group	N	$\bar{D}$ (°)	$\bar{I}$ (°)	R	α_{95}	Polarity
Young	12	21.5	62.5	11.78	6.2	Mixed (9N, 3R)
Old	22	29.5	65.5	21.07	6.7	Mixed (9N, 13R)

Note: Column headings as in Table 2. In this table Ohanapecosh and presumed Ohanapecosh rocks are separated into two groups on the basis of relative age. The means are shown to be substantially identical. For further discussion, see text.

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block appears to have rotated about 30° in a clockwise sense since late Eocene time.

Magill and others (1981) and Magill and Cox (1981) have advanced a two-phase model for the tectonic evolution of the Oregon Coast Range and Cascades. In the first phase, a seamount province formed on the Farallon plate is held to have rotated about a point near its southern end, by a method involving pivoting subduction that was first proposed by Menard (1978). Phase one, in this model, occurred during middle Eocene time and resulted in roughly 50° of clockwise rotation. In the second phase of the model, an additional 25° of clockwise rotation is believed to have resulted from differential expansion in the Basin and Range province. In this second phase the Klamath Mountains, the northern Sierra Nevada, and the Oregon Cascades moved as a unit. Phase two of Magill and others' (1981) model began in the early Miocene and may still be going on today. Crucial to this model, which builds on earlier analysis by Simpson and Cox (1977), is the presence of a hiatus in rotation lasting roughly from 40 to 20 m.y. B.P.—a little more than the entire Oligocene Epoch. Our data are compatible with this model but provide no direct test, because none of the Ohanapecosh rocks seem old enough to have been involved in the postulated first phase of rotation. Table 2 contains a hint of mid-Oligocene rotation, but, as mentioned above, the difference in magnetic declination between our "younger" and "older" groups is not significant statistically.

Beck and Plumley (1980) also have proposed a chronology of Coast Range rotation, based entirely on paleomagnetic data for the Oregon part of the range. The Beck and Plumley model, which proposes no specific driving mechanism, calls for steady rotation at roughly 1.5°/m.y. from the early Eocene until the early Miocene; events thereafter are regarded as indeterminant.

Unsurprisingly, we prefer the Beck and Plumley model. Its chief advantage is simplicity; we hold that, at the present state of investigation, the data do not offer sufficient constraint to justify a complicated model. Thus, we contend not that the Magill and others (1981) two-phase model is incorrect, but that it is unproved. We doubt that it is correct, however, for the following reasons.

1. The two-phase model calls for a hiatus in rotation during Oligocene time. However, three existing studies contain hints of Oligocene rotation. One, the study described here, concerns rocks adjacent to the Washington (not Oregon) Coast Range and thus may not be relevant (see below). However, the results of Beck and Plumley (1980) for Oregon Coast Range intrusive rocks and the directions reported by Beck (1962) for Western Cascade Series lavas in northern California both can be interpreted to support mid-Oligocene rotation. Unfortunately, none of these possible rotations are significant, in the usual sense of 95% confidence.

2. Several investigators, principally Magill and Cox (1981) and Beck and Plumley (1980), have contended that the Oregon and Washington Coast Ranges constitute two separate tectonic blocks. The basis for this interpretation can be seen in Figure 1. Oregon Coast Range studies all show rotations exceeding 40°; these are consistent from place to place and can be shown to decrease with age. The Oregon Coast Range thus appears to have rotated as a coherent block. The Washington Coast Range (including exposures on Vancouver Island), by contrast, shows rotations that are emphatically not consistent from place to place and that range from about 40° clockwise to perhaps 20° counterclockwise. Perhaps the Washington Coast Range consists of several small blocks that rotated independently upon accretion (Beck, 1980, suggested several mechanisms). Alternatively, if the Washington Coast Range were once a coherent block, it might have been severely distorted by underthrusting accompanying orogeny in the Olympic Mountains (Beck and Engebretson, 1980). In either event, using data from the Washington Coast Range to constrain rotation in the Oregon Coast Range seems unjustified. The hiatus in rotation during the Oligocene that is central to the two-phase model appears to be heavily constrained by two results (GV and OV, Table 1), both from Washington. We contend that had Magill and others (1981) performed their calculations on Oregon Coast Range data only, their two-phase model would have merged with our simpler scheme. This would be especially likely if they also had excluded the results of Clark (1969) for the Marys Peak sill, which is simply an intensively sampled single Oligocene intrusion that cannot have averaged out the geomagnetic secular variation. Figure 6, showing alternative rotational schemes, illustrates the basis of our contention. We emphasize that we have no quarrel with the mechanisms of rotation proposed by Magill and others (1981) and Magill and Cox (1981). Very likely, both rotation during oblique subduction and rotation by oblique rifting were involved, with the former necessarily preceding the latter. We maintain, however, that elements of timing, as well as details of the exact mechanisms, remain to be worked out.

The mean inclination of our Ohanapecosh rocks also deserves some scrutiny. As shown in Table 2, it is 63.5°, only 1.5° less than the expected inclination. In terms of the "flattening" statistic of Beck (1980), the Ohanapecosh inclination is flattened by 1.5° ± 9.0°—far from significant, at the 95% confidence level. This follows the general pattern for the Pacific Northwest (Beck, 1980); of all Tertiary rocks from the Cascades and Coast Ranges studied so far, only the Goble Volcanics (GV; Beck and Burr, 1979) have an inclination that is discordant, in the sense

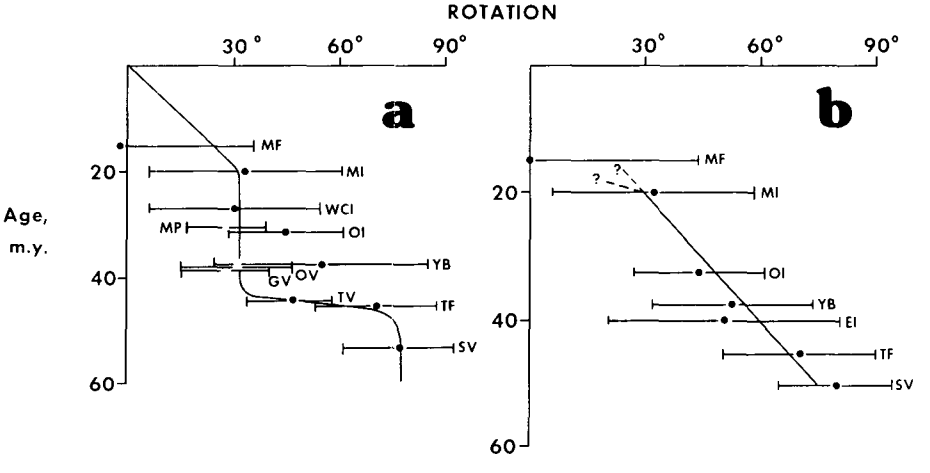


Figure 6. Alternative rotation schemes. (a) Modified from Magill and others (1981); (b) from Beck and Plumley (1980). Determinations shown as open circles are discussed in text. Letter symbols as in Figure 1 and Table 1.

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that its flattening exceeds the appropriate 95% confidence interval. However, closer examination of the data (tables in Beck, 1980, and Magill and others, 1981, are helpful) indicates the case is not this simple; of 15 paleomagnetic studies on Paleogene rocks from Oregon, Washington, and southern British Columbia known to us, only three have inclinations that are too steep. The exceptions are two studies of old Cascade volcanic rocks from the extreme southern end of the range (WC1, WC2; Beck, 1962; Magill and Cox, 1981) and a preliminary study of volcanic rocks associated with the Republic graben, in northeastern Washington (SP; Schwimmer and others, 1979). The prevalence of positive flattening values is far too marked to attribute to chance. As shown by M. E. Beck, Jr., and others (in prep.) for the case of the Clarno Formation of north-central Oregon, the choice of reference pole makes little difference. A persistent "far-sidedness" due to some zonal asymmetry of the geomagnetic field, such as those postulated recently by Coupland and Van der Voo (1980), also might account for the predominance of flattened inclinations. However, it seems far more likely that the flattened inclinations actually reflect a small amount of northward transport—enough to bias the table of inclinations toward values generally lower than expected, but not enough to provide the roughly 800 km of northward transport needed to produce flattenings that are significant statistically (Beck, 1980). Northward transport could include movement as part of the Farallon plate before accretion, or displacement by right-lateral faulting after accretion, or both. We further suspect that the negative flattening values encountered in the southern Cascade rocks reflect a real difference in tectonic history.

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